

RESEARCH ARTICLE

Exploration of Nonsuicidal Self-Injury as an Addiction-Like Behaviour in Depressed Adolescents in the light of the I-PACE Model

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ABSTRACT

Nonsuicidal self-injury (NSSI) is increasingly conceptualized as an addiction-like behaviour characterized by dysregulated emotional and cognitive processes. Guided by the I-PACE model, this study investigated how person-level vulnerabilities interact with affective, mental and executive functioning to maintain NSSI in clinically depressed adolescents ($N = 167$, aged 12–18, $M = 15.37 \pm 1.75$ years). Results revealed strong addiction-like patterns. Childhood trauma, depression and rumination demonstrated significant associations with NSSI frequency ($r = 0.59$ – 0.61), while resilience and self-esteem served as protective factors ($r = -0.53$ to -0.55). A hierarchical regression model explained 69% of variance, with trauma ($OR = 1.12$), depressive severity ($OR = 1.11$), rumination ($OR = 1.11$) and resilience ($OR = 0.90$) emerging as key predictors. Mediation analyses demonstrated how these factors operate in the addictive chain. Childhood trauma and borderline traits lead to affective dysregulation, which drives cognitive deficits that ultimately undermine resilience and increase NSSI risk ($\beta = -0.28$ and -0.24). These findings support the use of an addiction framework to conceptualize NSSI, while highlighting resilience-focused interventions as critical for breaking these maladaptive cycles.

1 | Introduction

The intentional, direct destruction of one's body tissue without suicidal intent is known as nonsuicidal self-injury (NSSI) (Klonsky 2011). Cutting, burning, scratching and other self-injurious behaviours are prevalent practices often used to alleviate severe psychological distress. It has been consistently linked to an increased risk of future suicide attempts and significant

psychiatric comorbidity, including depression and borderline personality disorder (Bentley et al. 2015; Fan et al. 2021). Given its clinical urgency, the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) classifies NSSI as a condition requiring further study.

With incidence rates ranging from 17% to 29% globally, epidemiological research confirms that adolescents are especially

Summary

- Early identification of person-level vulnerabilities, such as childhood trauma and borderline personality features, is critical for preventing the development and escalation of NSSI.
- Affective (e.g., depressive symptoms), cognitive (e.g., rumination, low self-esteem) and executive (e.g., poor resilience) dysfunctions interact to sustain self-injurious behaviour, highlighting the need for domain-specific assessments in clinical settings.
- Interventions that focus on strengthening resilience, enhancing emotional regulation and targeting maladaptive cognitive processes may disrupt the compulsive NSSI cycle and improve treatment outcomes.
- The I-PACE model offers a valuable, empirically supported framework for developing individualized, multilevel therapeutic strategies tailored to the psychological profiles of adolescents engaging in NSSI.

susceptible to NSSI (Xiao et al. 2022). In China, the 12-month prevalence of NSSI among teenagers is projected to be 29.2%, with proportions considerably higher among clinical groups with mood disorders (Liang et al. 2022). Najmi et al. (2007) also reported that Chinese teenagers suppress their emotions more than American adolescents, potentially increasing their risk of self-harm. Given the cultural tendency towards emotional restraint, it is imperative to investigate how

multiple factors influence the occurrence and severity of NSSI in Chinese youth.

Traditionally, NSSI has long been regarded as a maladaptive emotion regulation strategy associated with difficulties managing affect (Adrian et al. 2011), histories of childhood maltreatment (Martin et al. 2017) and cognitive vulnerabilities (Hasking et al. 2017). Emerging evidence, however, views NSSI through the lens of behavioural addiction. Individuals who self-injure frequently report cravings, escalating tension before the act and relief afterward (Chloe 2015), mirroring patterns seen in gambling or internet gaming disorders (Blasco-Fontecilla et al. 2016; Brandtner et al. 2021). NSSI's cyclic occurrence, as shown in Figure 1, is sustained by negative reinforcement rather than reward-seeking, distinguishing it from classic addictions. The four-function model of NSSI (Nock and Cha 2009) supports this cycle, highlighting how both positive (emotional relief) and negative (distraction from pain) reinforcement maintain the behaviour. Recent studies focus on the addiction-like nature of NSSI, with adolescents exhibiting compulsive withdrawal symptoms and persistent engagement despite adverse consequences (Zhu et al. 2022). This repetitive pattern aligns with behavioural addiction models, where repetitive acts become habits through operant conditioning (Robbins and Clark 2015). Experts are now exploring theoretical frameworks that have been employed in the study of behavioural addictions.

One of the latest models, the Interaction of Person-Affect-Cognition-Execution (I-PACE) model, introduced by Brand et al. (2019), is widely used in exploring addictive behaviours

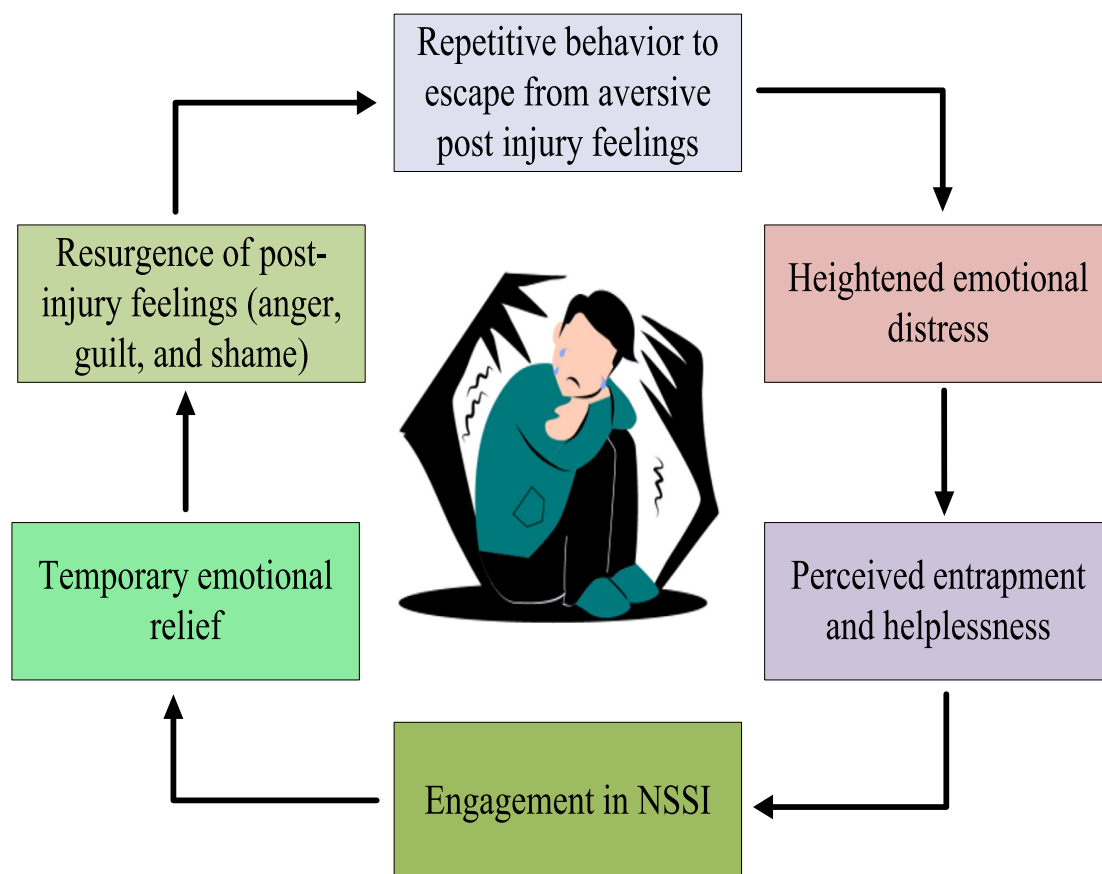


FIGURE 1 | Negative reinforcement drives NSSI occurrence in the cycle.

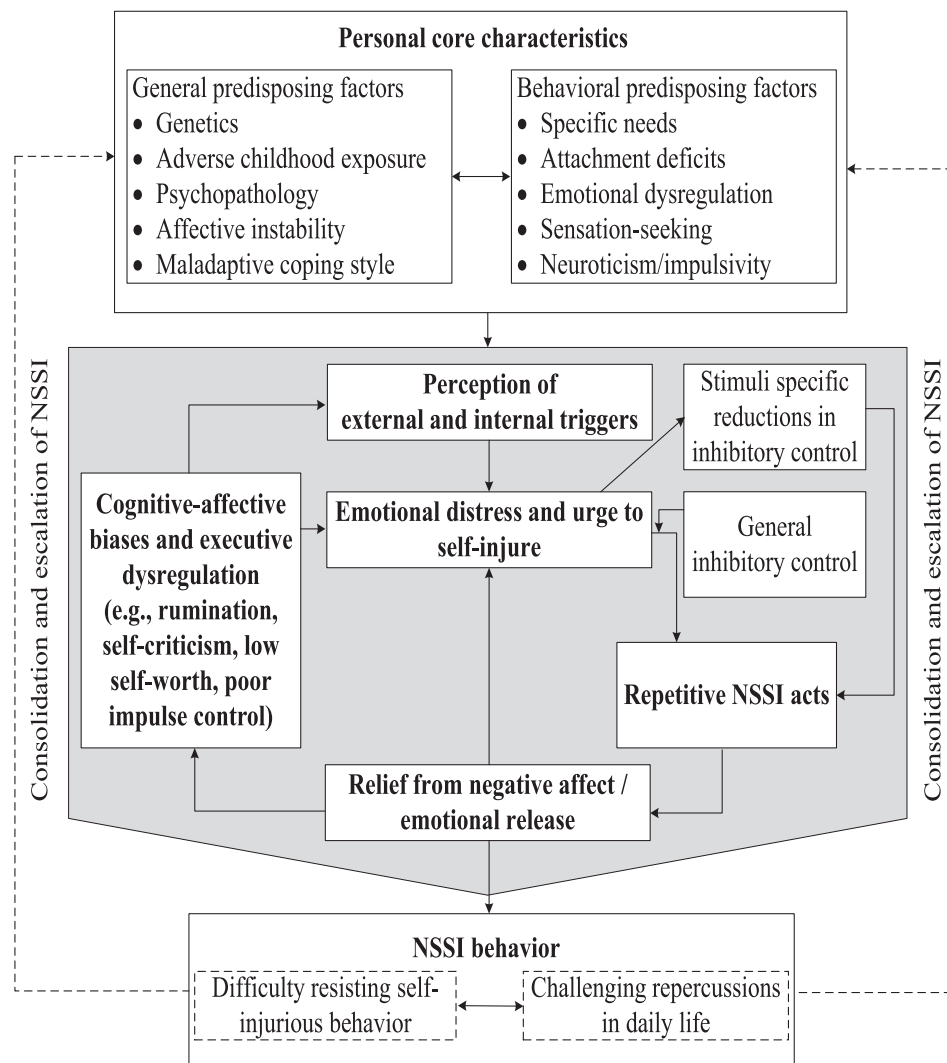


FIGURE 2 | I-PACE model as a conceptual framework in the context of NSSI.

(Mehmood et al. 2021; Stanciu and Calugar 2022). This model proposes an engaging and comprehensive approach to compulsive behaviours. Originally developed to explain Internet-related disorders, the I-PACE model states that addictive disorders stem from dynamic interactions among four domains: person-related vulnerabilities, affective responses, cognitive processes and executive-level capacities. Figure 2 is an explanation of the I-PACE model in the context of NSSI, illustrating how multiple factors interact in NSSI. The first study by Liu et al. (2022) validated the I-PACE model for evaluating NSSI as an addictive behaviour. The bulk of the existing research focuses on individual variables without considering their interdependence. Childhood trauma and borderline traits reliably predict both the occurrence and frequency of self-harm in adolescents (Ghinea et al. 2021; Zashchirinskaia and Isagulova 2023). Depressive symptoms and difficulties with emotion regulation are consistently associated with NSSI engagement (Chen et al. 2023; Faura-Garcia et al. 2024). Rumination affects the NSSI cycle, while low self-esteem also accelerates self-injurious behaviour (Buecker et al. 2025; Coleman et al. 2022). At the execution level, impulsivity and a passive coping style are proximal risk factors (Ma et al. 2018), while higher resilience decelerates the effects

of psychological distress and reduces self-injurious tendencies (Weedage et al. 2025). For adolescents whose prefrontal cortex is still developing, executive functions are crucial for managing the effects of emotional fluctuations and environmental stress (Friedman and Robbins 2022).

Despite the well-documented relationships between various factors and NSSI, the novelty of our study lies in its systematic application of the entire I-PACE model to NSSI. The present study addresses this gap by applying the I-PACE model to a clinical adolescent sample, aiming to unpack how personal vulnerabilities, affective instability, cognitive biases and executive functioning deficits jointly sustain NSSI as an addiction-like behaviour. This approach not only enriches theoretical understanding but also offers real-world implications. Findings from this study may inform the development of early screening tools, school-based prevention programmes focused on emotional regulation and resilience, and digital interventions that target specific domains of dysfunction highlighted by the I-PACE model.

The present study addresses a critical gap by examining the I-PACE model as a conceptual framework to elucidate the mechanistic pathways linking person-level vulnerabilities to

NSSI frequency among adolescents. This approach aims to clarify how components of the I-PACE model amplify the risk of NSSI.

Consistent with these aims, we propose the following hypotheses:

Hypothesis 1. *Person-level vulnerabilities (i.e., childhood trauma and borderline personality traits) will be significantly associated with increased NSSI engagement.*

Hypothesis 2. *Affective and cognitive dysfunctions (e.g., depressive symptoms, emotion dysregulation, rumination, low self-esteem) will mediate the relationship between person-level vulnerabilities and NSSI severity.*

Hypothesis 3. *Executive-level resources (i.e., resilience) will serve as a protective factor, reducing the likelihood of NSSI and weakening the indirect effects of person- and affect-level variables.*

It is anticipated that it will not only enhance the mechanistic understanding of the underlying psychological processes contributing to NSSI but also pave the way for the development of more targeted, domain-specific intervention strategies, offering hope for effective NSSI management.

2 | Methods

2.1 | Participants

For this study, 167 adolescents (69 males and 98 females) aged 12–18 years who had been diagnosed with major depressive disorder according to DSM-5 were selected by qualified psychiatrists at the Department of Depression, Shenzhen Kangning Hospital, Shenzhen, China. Participants were selected according to predefined inclusion and exclusion criteria, and written informed consent was obtained from each participant's parent or legal guardian. All data were anonymized, and participants were assured that their confidentiality would be maintained and that their participation would not affect their current or future treatment. The Institutional Review Board of Shenzhen Kangning Hospital ethically approved the study (approval#2020-K021-04-1).

The inclusion criteria were as follows: (1) NSSI behaviour within the past year, (2) no severe psychiatric diagnoses other than MDD (e.g., schizophrenia and bipolar disorder), (3) no history of drug use and (4) no physical disability or severe medical illness. Exclusion criteria included the following: (1) no NSSI behaviour in the past year, (2) the presence of neurological disorders or significant cognitive impairment and (3) current substance abuse.

2.2 | Psychological Measures

Standardized and validated Chinese versions of established instruments assessed psychological and behavioural factors consistent with the I-PACE model. Demographic variables, including age, gender and educational level, were also recorded. The following instruments were administered:

2.2.1 | Functional Assessment of Self-Mutilation (FASM)

Participants' NSSI behaviour was assessed using the FASM Chinese version (Qu et al. 2022). The FASM evaluated the presence, frequency and functional purpose of 10 types of NSSI behaviours that occurred in the past 12 months. Each behaviour was rated on a 5-point Likert scale. The total scores provided an overall frequency index, with higher totals indicating more frequent self-harm. Participants were categorized into two groups: (1) NSSI-positive ($n=46$) and (2) NSSI-negative ($n=43$) based on scoring. This scale demonstrated high reliability in the present sample (Cronbach's $\alpha=0.879$). Example item is 'How often have you cut yourself on purpose in the past 12 months?' Higher scores reflect more frequent engagement in NSSI behaviours.

2.2.2 | Childhood Trauma Questionnaire (CTQ)—Personal Component

The Chinese version of the CTQ validated by Zhao et al. (2005) is a retrospective questionnaire consisting of 28 items that assess five types of childhood maltreatment: emotional abuse, physical abuse, sexual abuse, emotional neglect and physical neglect, using a Likert scale from 1 = *never true* to 5 = *very true* (Zhao et al. 2005). Each subscale generates scores ranging from 5 to 25, with the total score varying from 25 to 125. In our sample, this scale demonstrated good reliability ($\alpha=0.891$), confirming its appropriateness for evaluating experiences of maltreatment in depressed adolescents. Example item is 'People in my family hit me so hard that it left me with bruises or marks'. Higher scores indicate greater exposure to childhood abuse or neglect.

2.2.3 | Borderline Personality Features Scale for Children (BPFS-C)—Personal Component

The Chinese version of the BPFS-C comprises 24 items that measure core borderline personality traits, including emotional instability, impulsivity and identity disruption (Zhuo et al. 2022). Participants respond using a five-point Likert-type scale, resulting in a total score ranging from 24 to 120, with subscale scores varying from 6 to 30. In our sample of depressed adolescents, the scale exhibited excellent reliability ($\alpha=0.862$). Example item is 'I feel very angry a lot of the time'. Higher scores indicate greater severity of borderline personality traits such as affective instability and impulsivity.

2.2.4 | Beck Depression Inventory-II (BDI-II)—Affective Component

The Chinese version of the BDI-II is a 21-item self-report scale that assesses the severity of depression symptoms experienced over the past 2 weeks (Wang and Gorenstein 2013). Each item is rated on a 4-point scale, ranging from 0 to 3, yielding total scores between 0 and 63. Each item represents a specific symptom of depression, such as sadness, guilt, fatigue or changes in sleep or appetite. In the current study, the BDI-II demonstrated excellent

reliability ($\alpha=0.911$). Example item is 'I feel sad all the time, and I can't snap out of it'. Higher scores represent more severe depressive symptoms.

2.2.5 | Emotion Regulation Questionnaire for Children and Adolescents (ERQ-CA)—Affective Component

The Chinese version of ERQ-CA is a 10-item self-report instrument modified from the original ERQ. It is used to assess two primary emotion regulation strategies in adolescents: expressive suppression (four items) and cognitive reappraisal (six items) (Liu et al. 2017). Each item is rated by respondents on a 7-point Likert scale, where one represents *strongly disagree* and seven signifies *strongly agree*. The scale has shown satisfactory internal consistency ($\alpha=0.835$). Examples are (cognitive reappraisal): 'I control my emotions by changing the way I think about the situation I'm in', and (expressive suppression): 'I keep my emotions to myself'. Higher reappraisal scores reflect adaptive emotion regulation; higher suppression scores may indicate less adaptive strategies.

2.2.6 | Ruminative Responses Scale (RRS)—Cognitive Component

The RRS Chinese version is an instrument comprising 22 items that can be used to measure the frequency of rumination as a reaction to emotional stress in three dimensions: symptom rumination, rumination and reflective rumination (Yang et al. 2020). Respondents rate each statement on a 4-point scale (1 = *never* to 4 = *always*), and mean scores are calculated, with higher scores indicating a greater tendency to ruminate. The scale has shown excellent internal consistency in adolescent samples (Cronbach's $\alpha=0.93$). Example item is 'I think about how alone I feel'. Higher scores indicate a stronger tendency to ruminate in response to emotional distress.

2.2.7 | Rosenberg Self-Esteem Scale (RSES)—Cognitive Component

The RSES, a Chinese adaptation, consists of 10 items designed to assess self-esteem and self-worth. Five items are reverse-scored, and each item is evaluated on a 4-point scale (Jiang et al. 2023). Higher scores indicate better self-esteem; total scores range from 10 to 40, and this scale demonstrated good reliability ($\alpha=0.881$) in the present study. Example item is 'I feel that I have a number of good qualities'. Higher scores suggest greater self-esteem and self-worth.

2.2.8 | Connor–Davidson Resilience Scale (CD-RISC)—Executive Component as Adaptive Coping

The CD-RISC scale, adapted by Yu and Zhang (2007), was employed to evaluate executive function-based adaptive coping abilities. This 10-item scale measures three dimensions important for stress-related behavioural regulation: (1) tenacity (persisting despite adversity), (2) strength (rapid recovery from distress) and (3) optimism (a positive future orientation).

Respondents scored items on a 5-point scale (1 = *never*, 5 = *always*), with higher scores indicating stronger resilience. In our sample, the scale demonstrated excellent internal consistency ($\alpha=0.883$). Example item is 'I am able to adapt when changes occur'. Higher scores indicate stronger resilience and adaptive coping capacity.

The CD-RISC scale was employed to assess the executive component of adaptive coping in the I-PACE model, following its established use in prior research examining executive functioning (Southwick and Charney 2012). This selection was justified by the scale's tenacity, strength and optimism, aligning with key executive processes, including goal-directed persistence, emotional regulation and cognitive reappraisal (Zhou et al. 2024).

2.3 | Statistical Analysis

Before data analysis, we employed the common latent factor (CLF) approach to address any common method variance (CMV) because of the behavioural nature of the data (Podsakoff et al. 2003). Demographic differences between NSSI groups were investigated using independent-sample *t* tests (age) and chi-square tests (gender and education). Correlations were used to measure the direction and strength of associations among all domain scores and their relationships with dichotomous NSSI status. False discovery rate (FDR) correction was applied.

We used hierarchical logistic regression to determine the contribution of each domain beyond earlier phases. Gender and age were taken into account as confounding variables in all analyses prior to adding any of the other variables (block 0). SPSS, PROCESS, Model 6 (v3.5), with 5000 bias-corrected bootstrap samples, was used to confirm the chain mediating effects of I-PACE components on the frequency of NSSI. Childhood trauma (CTQ) or borderline traits (BPFS-C) were taken as the independent variable, with affect (BDI-II, ERQ-CA), cognition (RRS, RSES) and execution (CD-RISC) entered sequentially as mediators, and NSSI as the outcome. We included age and sex as covariates in all paths. Indirect effects were significant, with the 95% bootstrap CI not including zero.

3 | Results

3.1 | Assessment of Standard Method Variance

Because our data were self-reported, we employed the CLF technique (Podsakoff et al. 2003) to rigorously evaluate any potential common method bias. This confirmatory approach compared two models: (1) the original measurement model and (2) a bifactor model with an additional method factor that accounts for item covariance. The analysis's findings, as shown in Table 1, indicate that there are few differences between the models. Fit indices for the bifactor model and the measurement model differed very slightly ($\Delta CFI=0.006$, $\Delta TLI=0.006$, $\Delta SRMR=-0.003$, $\Delta RMSEA=0.004$). Notably, the change in CFI was below the 0.01 threshold suggested for meaningful differences. The χ^2/df ratio showed modest improvement (from 1.969 to 1.813). These show that CMV is not a threat to the validity of our findings,

and respondents were able to distinguish between the substantive constructs in our study.

3.2 | Comparison of Demographic Characteristic

The demographic characteristics of NSSI-positive and NSSI-negative subjects were compared as shown in Table 2. There were no significant differences between the two groups in terms of age ($t = -1.46$, $p = 0.148$) or years of education ($t = -1.69$,

$p = 0.094$). In contrast, there was a greater proportion of males in the NSSI-negative group (51.28%) than in the NSSI-positive group (32.58%), reflecting a significant difference in the sex distribution ($\chi^2 = 15.22$, $p = 0.001$).

3.3 | Strength and Direction of Associations Between I-PACE Measures and NSSI

Table 3 displays intercorrelation among I-PACE model components and NSSI. Childhood trauma was positively associated with depressive symptoms ($r = 0.58$, $p < 0.01$), rumination ($r = 0.41$, $p < 0.01$), and NSSI ($r = 0.59$, $p < 0.01$). Borderline personality features were also significantly correlated with rumination ($r = 0.52$, $p < 0.01$) and NSSI ($r = 0.36$, $p < 0.01$). Depression was strongly associated with rumination ($r = 0.53$, $p < 0.01$) and NSSI ($r = 0.61$, $p < 0.01$). Cognitive reappraisal was negatively associated with NSSI ($r = -0.32$, $p < 0.01$), while expressive suppression showed a small but significant positive correlation ($r = 0.38$, $p < 0.05$). Rumination was significantly related to lower resilience ($r = -0.54$, $p < 0.01$) and greater NSSI engagement ($r = 0.42$, $p < 0.01$). Furthermore, self-esteem and resilience were both negatively correlated with NSSI ($r = -0.53$ and -0.55 , respectively; $p < 0.01$ for both). These patterns are consistent with theoretical expectations derived from the I-PACE model.

3.4 | NSSI Risk Factors Based on the I-PACE Model

The determining factor of NSSI within the I-PACE framework was evaluated using hierarchical logistic regression, facilitating the assessment of the distinct and combined effects of key psychological domains, as shown in Table 4. The model progressed through blocks that represented the person, affective, cognitive and executive domains. In Step 1 (person domain), borderline personality traits ($OR = 1.09$ [1.02, 1.17], $p = 0.011$) and childhood trauma ($OR = 1.12$, 95% CI [1.04, 1.21], $p = 0.003$) turned out to be significant predictors, accounting for 38% of the variance in NSSI status ($R^2 = 0.38$, $\chi^2(2) = 25.1$, $p < 0.001$).

TABLE 1 | Common latent factor analysis for detecting potential CMV.

Model	χ^2/df	CFI	TLI	SRMR	RMSEA
Measurement model	1.969	0.949	0.952	0.031	0.042
Bifactor model	1.813	0.943	0.958	0.034	0.046
Difference (Δ)	—	0.006	0.006	-0.003	0.004

Note: χ^2/df = chi-square to degrees of freedom ratio; CFI = comparative fit index; TLI = Tucker-Lewis's index; SRMR = standardized root mean square residual; RMSEA = root mean square error of approximation.

TABLE 2 | Demographic characteristics comparison between NSSI-positive and NSSI-negative groups.

Variable	NSSI-positive ($n = 89$) n (%)	NSSI-negative ($n = 78$) n (%)	t/χ^2	p
Age	15.13 (1.86)	15.67 (1.66)	-1.46	0.148 ^a
Education (years)	9.39 (2.26)	10.12 (1.78)	-1.69	0.094 ^a
Sex (male)	29 (32.58%)	40 (51.28%)	15.22	0.001 ^{b*}
Sex (female)	60 (67.42%)	38 (48.72%)		

Note: a: Independent samples t tests were used to compare means of continuous variables; b: Chi-square tests (χ^2) were used for categorical variables (sex).

* $p < 0.05$ indicates statistical significance.

TABLE 3 | Intercorrelations among I-PACE measures and NSSI.

Variable	1	2	3	4	5	6	7	8	9
1. CTQ	—								
2. BPFS-C	0.43**	—							
3. BDI-II	0.58**	0.35**	—						
4. ERQ-cognitive reappraisal	0.32**	0.29*	0.47**	—					
5. ERQ-expressive suppression	0.10*	0.08*	0.12*	0.12*	—				
6. RRS	0.41**	0.52**	0.53**	0.42**	0.20**	—			
7. RSES	0.45**	0.41**	0.48**	0.36**	-0.14**	0.58**	—		
8. CD-RISC	-0.42**	-0.39**	-0.46**	-0.30**	-0.20**	-0.54**	-0.50**	—	
9. NSSI	0.59**	0.36**	0.61**	-0.32**	0.38*	0.42**	-0.53**	-0.55**	—

Note: * $p < 0.05$; ** $p < 0.01$.

TABLE 4 | Hierarchical logistic regression analysis predicting NSSI.

Domain	Predictor	B	SE	OR	95% CI	<i>p</i>	Nagel. R^2	χ^2
1. Person							0.38	25.1***
	CTQ	0.11	0.04	1.12	[1.04, 1.21]	0.003		
	BPFS-C	0.09	0.04	1.09	[1.02, 1.17]	0.011		
2. +Affect							0.16	14.3***
	BDI-II	0.10	0.05	1.11	[1.01, 1.22]	0.028		
	ERQ-CA	0.06	0.03	1.06	[1.00, 1.13]	0.049		
3. +Cognition							0.10	12.1**
	RRS	0.10	0.04	1.11	[1.03, 1.20]	0.007		
	RSES	−0.12	0.01	0.89	[1.00, 1.05]	0.041		
4. +Execution							0.05	7.6**
	CD-RISC	−0.10	0.03	0.90	[0.84, 0.96]	0.003		

Note: *** $p < 0.001$; ** $p < 0.01$; odds ratio (OR) represents the change in odds of NSSI for a one-unit increase in the predictor; Nagelkerke R^2 indicates the proportion of variance explained by each step in the model.

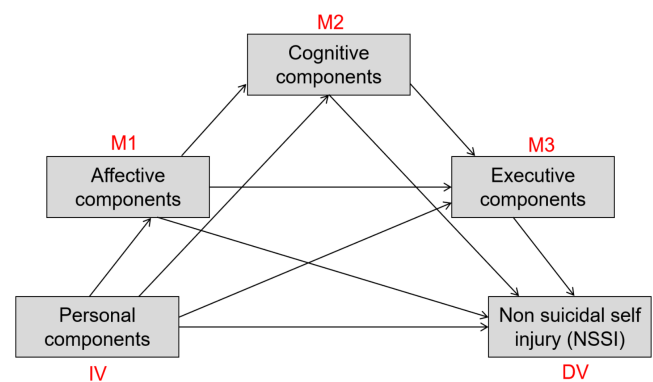
Incorporating Step 2 (affect domain) variables improved model fit significantly ($\Delta R^2 = 0.16$, $\Delta\chi^2(2) = 14.3$, $p < 0.001$), with depression severity (OR = 1.11 [1.01, 1.22], $p = 0.028$) and emotion dysregulation (OR = 1.06 [1.00, 1.13], $p = 0.049$) contributing independently. The percentage of variance explained improved to 54%. In Step 3 (cognition domain), predictive accuracy was further enhanced ($\Delta R^2 = 0.10$, $\Delta\chi^2(2) = 12.1$, $p = 0.002$), with rumination (OR = 1.11 [1.03, 1.20], $p = 0.007$) and self-esteem (OR = 0.89 [1.00, 1.05], $p = 0.041$) as significant predictors, bringing the cumulative explained variance to 64%. In Step 4 (execution domain), resilience was found to be a protective factor (OR = 0.90 [0.84, 0.96], $p = 0.003$), leading to a modest but significant improvement ($\Delta R = 0.05$, $\Delta\chi = 7.6$, $p = 0.006$). All domains contributed much more than those in the previous steps, and the final model explained 69% of the variance in NSSI status.

3.5 | Detection of I-PACE Component as Mediators

A chain mediation analysis was conducted to examine the role of multiple mediators in the relationship between person-level components and NSSI. The conceptual mediation model is illustrated in Figure 3 to facilitate quick comprehension.

3.5.1 | Model 1 (CTQ as Predictor)

To investigate whether the effect of childhood trauma (CTQ) on NSSI severity is mediated by a sequential pathway involving affective, cognitive and executive mechanisms, a chain mediation analysis was conducted with 5000 bootstrap samples. Results in Table 5 showed that higher CTQ scores were significantly associated with increased affective dysregulation (BDI-II/ERQ-CA) ($\beta = 0.85$, SE = 0.12, $p < 0.001$, 95% CI [0.61, 1.09]). Greater affective dysregulation, in turn, predicted increased cognitive difficulties (RRS/RSES) ($\beta = 0.78$, SE = 0.14, $p < 0.001$, 95% CI [0.51, 1.04]). These cognitive difficulties subsequently predicted lower levels of resilience

**FIGURE 3** | Study conceptual model: Personal components with multiple mediators and NSSI as an outcome in a series mediation model.

(CD-RISC) ($\beta = -0.65$, SE = 0.15, $p < 0.001$, 95% CI [−0.94, −0.36]), which in turn was associated with increased NSSI severity ($\beta = -0.48$, SE = 0.11, $p < 0.001$, 95% CI [−0.69, −0.27]). Childhood trauma predicted NSSI severity through affective dysregulation, cognitive difficulties and reduced resilience ($B = -0.28$, 95% CI [−0.48, −0.12]). The negative coefficient indicates that lower resilience (execution) exacerbated NSSI severity. Although the direct effect remained significant ($B = 0.22$, $p = 0.009$), the full chain accounted for the most significant indirect effect. These findings support a sequential risk process in which early trauma contributes to emotional and cognitive impairments that undermine psychological resilience and thereby worsen NSSI severity.

3.5.2 | Model 2 (BPFS-C as Predictor)

A serial mediation analysis was performed to determine if borderline personality features (BPFS-C) affect the severity of NSSI through a stepwise pathway regarding affective, cognitive and executive functioning domains. Results in Table 6 showed that BPFS-C significantly predicted increased

TABLE 5 | Effect of CTQ on NSSI severity via affect, cognition and execution mediators (Model 1).

Path	Effect (β)	SE	<i>p</i>	95% CI
CTQ → affect (BDI-II/ERQ-CA)	0.85	0.12	<0.001	[0.61, 1.09]
Affect → cognition (RRS/RSES)	0.78	0.14	<0.001	[0.51, 1.04]
Cognition → execution (CD-RISC)	−0.65	0.15	<0.001	[−0.94, −0.36]
Execution → NSSI severity	−0.48	0.11	<0.001	[−0.69, −0.27]
Direct effect				
CTQ → NSSI severity	0.22	0.08	0.009	[0.06, 0.39]
Specific indirect effects				
CTQ → affect → NSSI severity	0.36	0.10		[0.18, 0.57]
CTQ → affect → cognition → NSSI severity	0.30	0.09		[0.13, 0.48]
CTQ → affect → cognition → execution → NSSI severity	−0.28	0.09		[−0.48, −0.12]
Total indirect effect	0.08	0.04		[0.01, 0.17]
Total effect (c)	0.30	0.09		[0.13, 0.48]

TABLE 6 | Effect of BPFS-C on NSSI severity via affect, cognition and execution mediators (Model 2).

Path	Effect (β)	SE	<i>p</i>	95% CI
BPFS-C → affect (BDI-II/ERQ-CA)	0.72	0.11	<0.001	[0.50, 0.93]
Affect → cognition (RRS/RSES)	0.78	0.14	<0.001	[0.51, 1.04]
Cognition → execution (CD-RISC)	−0.65	0.15	<0.001	[−0.94, −0.36]
Execution → NSSI severity	−0.48	0.11	<0.001	[−0.69, −0.27]
Direct effect				
BPFS-C → NSSI severity	0.18	0.07	0.011	[0.04, 0.32]
Specific indirect effects				
BPFS-C → affect → NSSI severity	0.31	0.08		[0.16, 0.49]
BPFS-C → affect → cognition → NSSI severity	0.20	0.06		[0.09, 0.33]
BPFS-C → affect → cognition → execution → NSSI severity	−0.24	0.07		[−0.40, −0.10]
Total indirect effect	0.07	0.03		[0.01, 0.15]
Total effect (c)	0.25	0.08		[0.10, 0.41]

affective dysregulation ($\beta=0.72$, $SE=0.11$, $p<0.001$, 95% CI [0.50, 0.93]), which in turn was associated with greater cognitive difficulties ($\beta=0.78$, $SE=0.14$, $p<0.001$, 95% CI [0.51, 1.04]). These cognitive challenges later on predicted reduced resilience ($\beta=-0.65$, $SE=0.15$, $p<0.001$, 95% CI [−0.94, −0.36]), which was coupled with elevated NSSI severity ($\beta=-0.48$, $SE=0.11$, $p<0.001$, 95% CI [−0.69, −0.27]). Even after accounting for these mediators, a direct effect of BPFS-C on NSSI severity remained significant ($\beta=0.18$, $SE=0.07$, $p=0.011$, 95% CI [0.04, 0.32]), thus declaring partial mediation through this serial process.

4 | Discussion

The present study employed the I-PACE model to conceptualize NSSI as an addictive behaviour and to examine its multilevel

psychological predictors. Our findings offer empirical support for this framework by demonstrating how individual vulnerabilities, affective dysregulation, maladaptive cognitive processes and compromised executive resources cumulatively increase the likelihood of engaging in NSSI.

A significant sex difference emerged in the distribution of NSSI, with a higher proportion of females in the NSSI-positive group. This finding is consistent with existing literature, which shows that adolescent females are more likely to engage in NSSI, potentially due to increased emotional dysregulation, internalizing symptoms, and sociocultural factors influencing coping behaviour (Bresin and Schoenleber 2015; Victor et al. 2018).

Our findings highlight the strong links between NSSI and several psychological variables, particularly childhood trauma

and depressive symptoms, while also identifying the protective roles of self-esteem and resilience. A strong association between childhood trauma and NSSI in our study echoes prior literature showing that early adversity impairs stress regulation and fosters emotion regulation deficits (Misiak et al. 2023; Zashchirinskaia and Isagulova 2023). NSSI may function as a form of self-punishment or a means of coping with negative feelings of worthlessness and self-criticism.

Consistent with Hypothesis 1, person-level vulnerabilities, particularly childhood trauma and borderline personality features emerged as significant predictors of NSSI. These traits may sensitize individuals to internal distress, increasing susceptibility to maladaptive coping strategies like NSSI. The researchers emphasized trauma as a foundational risk factor for NSSI (Glenn and Klonsky 2013). A substantial body of research also reported that early adverse experiences, especially emotional and physical abuse, are closely linked to self-injurious behaviour (Martin et al. 2017; Zashchirinskaia and Isagulova 2023). Aligning with our results, Chen et al. (2023) and Sadeh et al. (2014) also reported that borderline features were significantly associated with increased emotional instability and heightened risk of NSSI in depressed adolescents.

Supporting Hypothesis 2, depressive symptoms and difficulties in emotion regulation further elevated NSSI risk. This supports the model's assertion that emotional dysregulation serves as a key mechanism linking person-level vulnerabilities to compulsive behaviours. The strong associations observed between depression, emotional suppression and NSSI affirm prior findings that internalizing symptoms often precede and sustain self-injurious acts (Faura-Garcia et al. 2024; Ghinea et al. 2021). Adolescents with depression usually experience heightened psychological pain, hopelessness and self-directed anger that increase vulnerability to NSSI (Xavier et al. 2017). Emotion dysregulation was observed as an accelerator for NSSI, which is supported by a previous study that found emotional distress causes NSSI (Buelens et al. 2019). These findings are theoretically grounded in the affect regulation and experiential avoidance models, which propose that NSSI provides short-term emotional relief, thus reinforcing its use through negative reinforcement (Nock and Cha 2009).

We also found that cognitive dysfunctions, particularly rumination and low self-esteem, accelerated NSSI. Rumination was not only highly correlated with depression and borderline traits but also significantly predicted NSSI behaviour. Rumination, as suggested by Selby et al. (2013), is not only a risk factor but also has an 'addictive quality' that exacerbates emotional distress and promotes the repetition of self-injury. Another longitudinal study also reported similar findings regarding rumination and self-injury (Buelens et al. 2019). This is also consistent with cognitive models of NSSI, which view repetitive negative thinking as a cognitive trap that reduces behavioural flexibility and heightens self-harming urges (Nock and Cha 2009). Similarly, lower self-esteem also contributed to greater NSSI likelihood, suggesting a cognitive vulnerability that interacts with affective distress to erode coping capacity. Low self-esteem contributes to negative self-appraisals and heightened emotional distress, which individuals may attempt to alleviate through NSSI, as documented in prior research (Buecker et al. 2025; Lei

et al. 2024). This cognitive dysfunction is consistent with the I-PACE model, which posits that maladaptive cognitive patterns, heightened emotional reactivity and weakened behavioural control increase vulnerability to NSSI. It is also important to note that not all instances of NSSI arise from the same psychological motivations. For some adolescents, the behaviour may serve as a means of emotion regulation; for others, it may function as self-punishment or a form of communication. This motivational heterogeneity does not sabotage the I-PACE model but rather highlights its adaptability. The model's multidomain architecture accommodates different entry points and maintains conceptual flexibility across diverse NSSI profiles.

Confirming Hypothesis 3 at the execution level, high resilience significantly reduced the likelihood of NSSI, indicating a buffering effect. Resilience is defined as the capacity to adapt positively in the face of adversity (Weedage et al. 2025). Adolescents with higher resilience may possess stronger behavioural regulation and greater psychological flexibility, allowing them to resist the urge to self-injure even under emotional strain. Although not formally tested as a moderator, this result aligns with previous research that positions resilience as a safeguard against psychological distress and behavioural dysregulation (Southwick and Charney 2012; Zhang et al. 2023; Zhou et al. 2024). This buffering effect aligned with the I-PACE model's emphasis on executive control as a protective factor in the context of addictive behaviours.

These overall findings support the I-PACE model's utility in conceptualizing NSSI as an addiction-like behaviour driven by person-level vulnerabilities, affective dysregulation, maladaptive cognition and impaired executive control. Consistent with research on behavioural addictions and experiential avoidance theory, NSSI appears to function as a maladaptive coping strategy for regulating overwhelming negative emotions (Chapman et al. 2006). The model's integrative framework effectively captures the complexity of these interacting risk processes. Framing NSSI through an addiction model is not intended to replace aetiological approaches but rather to complement them by elucidating how distress may evolve into repetitive, compulsive behaviour patterns. This integrative lens enables multilevel interventions at both the root-cause level and the reinforcement-maintenance level.

4.1 | Implications

This study contributes to the growing literature by exploring a process-based understanding of NSSI. By applying the I-PACE model, we provide a structured, empirically grounded framework for identifying adolescents at risk. The model also holds value for preventive frameworks, offering clear entry points for school- or family-centred interventions. Clinical interventions should be multicomponent, targeting early trauma and personality vulnerabilities, improving emotional awareness, regulation and resilience skills and challenging maladaptive thought patterns.

Our findings support targeted interventions addressing specific I-PACE pathways: trauma-focused therapies for person-level vulnerabilities, dialectical behavioural therapy (DBT) skills

training for affective dysregulation, cognitive restructuring for rumination and low self-esteem and resilience training to strengthen executive functioning. A stepped-care approach ranging from psychoeducation for mild cases to intensive schema therapy for more severe presentations can be tailored to individual risk profiles.

Blended models combining therapist support with app-based tools (e.g., mood tracking) may optimize outcomes, particularly when culturally adapted to local emotion regulation norms. Monthly monitoring of resilience and rumination may help track progress across domains. Critically, fostering resilience and enhancing adaptive coping mechanisms are essential components of any intervention aimed at reducing NSSI. In educational settings, evidence-based resilience programmes can be effectively integrated through curriculum-embedded approaches. By helping adolescents develop healthier strategies for managing emotional distress, such interventions can reduce reliance on self-injury as a coping mechanism.

4.2 | Limitations and Future Directions

This study has several limitations. First, the reliance on self-report measures may introduce response biases, particularly in assessing emotion regulation and personality traits. Second, incorporating behavioural assessments and neuro-imaging techniques (fMRI/EEG) could offer deeper insights into the mechanisms underlying NSSI. A longitudinal design is also recommended to clarify temporal pathways and dynamic interactions within the I-PACE model. Third, executive functioning was not measured with dedicated scales. Our use of CD-RISC as an executive function proxy focused specifically on its self-regulatory aspects. Future studies should employ comprehensive scales to evaluate executive functioning. Additionally, the generalization of our findings is constrained by the relatively small and demographically homogeneous sample of Chinese adolescents diagnosed with MDD. Cultural attitudes towards mental health, emotion regulation and self-injurious behaviour can differ significantly across regions, potentially influencing both the expression of symptoms and the effectiveness of specific interventions. Future studies should recruit more diverse samples to examine whether the I-PACE pathways identified here hold across different cultural or clinical populations.

5 | Conclusion

This study advances the understanding of NSSI by applying the I-PACE model, which highlights the interplay between early vulnerabilities, emotional distress and maladaptive cognitive processes. The findings support conceptualizing NSSI as an addiction-like behaviour reinforced by depressive symptoms, rumination and low self-esteem. Conversely, psychological resources such as resilience and self-worth may disrupt this cycle by mitigating its reinforcing effects. Adolescents with predisposing vulnerabilities and impaired affective and cognitive control may resort to NSSI as a maladaptive coping mechanism. Targeted interventions, including mindfulness-based cognitive therapy, dialectical behaviour therapy and emotion-focused

approaches, could promote healthier coping strategies by strengthening resilience and emotional regulation, ultimately reducing reliance on self-injury.

Author Contributions

A.M. and Z.L. conceived and designed this study. S.X., L.Z. and S.M.S. collected the data. A.M. analyzed the data. A.M., S.X., L.Z., G.H. and Y.Z. drafted or revised the article. All authors gave final approval of the version to be published.

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Ethics Statement

The study protocol was ethically approved by the Institutional Review Board of Shenzhen Kangning Hospital (approval#2020-K021-04-1), and informed consent was obtained.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data used in this study will be available on reasonable demand through a valid official email from the corresponding author.

Declaration of Generative AI and AI-Assisted Technologies in the Writing

During the preparation of this work, the authors used the Grammarly tool to improve language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

- Adrian, M., J. Zeman, C. Erdley, L. Lisa, and L. Sim. 2011. "Emotional Dysregulation and Interpersonal Difficulties as Risk Factors for Nonsuicidal Self-Injury in Adolescent Girls." *Journal of Abnormal Child Psychology* 39: 389–400.
- Bentley, K. H., C. F. Cassiello-Robbins, L. Vittorio, S. Sauer-Zavala, and D. H. Barlow. 2015. "The Association Between Nonsuicidal Self-Injury and the Emotional Disorders: A Meta-Analytic Review." *Clinical Psychology Review* 37: 72–88.
- Blasco-Fontecilla, H., R. Fernández-Fernández, L. Colino, L. Fajardo, R. Perteguer-Barrio, and J. De Leon. 2016. "The Addictive Model of Self-Harming (Non-Suicidal and Suicidal) Behavior." *Frontiers in Psychiatry* 7, no. 8.
- Brand, M., E. Wegmann, R. Stark, et al. 2019. "The Interaction of Person-Affect-Cognition-Execution (I-PACE) Model for Addictive Behaviors: Update, Generalization to Addictive Behaviors Beyond Internet-Use Disorders, and Specification of the Process Character of Addictive Behaviors." *Neuroscience & Biobehavioral Reviews* 104: 1–10.

- Brandtner, A., S. Antons, A. Cornil, and M. Brand. 2021. "Integrating Desire Thinking Into the I-PACE Model: A Special Focus on Internet-Use Disorders." *Current Addiction Reports* 8: 459–468.
- Bresin, K., and M. Schoenleber. 2015. "Gender Differences in the Prevalence of Nonsuicidal Self-Injury: A Meta-Analysis." *Clinical Psychology Review* 38: 55–64.
- Buecker, S., J. Doll, S. Abrantes Diaz, et al. 2025. "Self-Esteem and Suicidal Thoughts and Behaviors: A Meta-Analytic Review." *Clinical Psychological Science*: 21677026241308163.
- Buelens, T., K. Luyckx, A. Gandhi, G. Kiekens, and L. Claes. 2019. "Non-Suicidal Self-Injury in Adolescence: Longitudinal Associations With Psychological Distress and Rumination." *Journal of Abnormal Child Psychology* 47: 1569–1581.
- Chapman, A. L., K. L. Gratz, and M. Z. Brown. 2006. "Solving the Puzzle of Deliberate Self-Harm: The Experiential Avoidance Model." *Behaviour Research and Therapy* 44, no. 3: 371–394.
- Chen, Y., W. Fu, S. Ji, et al. 2023. "Relationship Between Borderline Personality Features, Emotion Regulation, and Non-Suicidal Self-Injury in Depressed Adolescents: A Cross-Sectional Study." *BMC Psychiatry* 23, no. 1: 293.
- Chloe, H. 2015. "Nonsuicidal Self-Injury and Suicidal Risk: An Examination Among Young Adults".
- Coleman, S. E., B. J. Dunlop, S. Hartley, and P. J. Taylor. 2022. "The Relationship Between Rumination and NSSI: A Systematic Review and Meta-Analysis." *British Journal of Clinical Psychology* 61, no. 2: 405–443.
- Fan, Y.-y., J. Liu, Y.-y. Zeng, R. Conrad, and Y.-l. Tang. 2021. "Factors Associated With Non-Suicidal Self-Injury in Chinese Adolescents: A Meta-Analysis." *Frontiers in Psychiatry* 12: 747031.
- Faura-Garcia, J., E. Calvete, and I. Orue. 2024. "Longitudinal Associations Between Nonsuicidal Self-Injury, Depressive Symptoms, Hopelessness, and Emotional Dysregulation in Adolescents." *Archives of Suicide Research* 28, no. 3: 800–814.
- Friedman, N. P., and T. W. Robbins. 2022. "The Role of the Prefrontal Cortex in Cognitive Control and Executive Function." *Neuropsychopharmacology* 47, no. 1: 72–89.
- Ghinea, D., A. Fuchs, P. Parzer, J. Koenig, F. Resch, and M. Kaess. 2021. "Psychosocial Functioning in Adolescents With Non-Suicidal Self-Injury: The Roles of Childhood Maltreatment, Borderline Personality Disorder, and Depression." *Borderline Personality Disorder and Emotion Dysregulation* 8: 1–11.
- Glenn, C. R., and E. D. Klonsky. 2013. "Nonsuicidal Self-Injury Disorder: An Empirical Investigation in Adolescent Psychiatric Patients." *Journal of Clinical Child & Adolescent Psychology* 42, no. 4: 496–507.
- Hasking, P., J. Whitlock, D. Voon, and A. Rose. 2017. "A Cognitive-Emotional Model of NSSI: Using Emotion Regulation and Cognitive Processes to Explain why People Self-Injure." *Cognition and Emotion* 31, no. 8: 1543–1556.
- Jiang, C., Y. Zhu, Y. Luo, et al. 2023. "Validation of the Chinese Version of the Rosenberg Self-Esteem Scale: Evidence From a Three-Wave Longitudinal Study." *BMC Psychology* 11, no. 1: 345.
- Klonsky, E. D. 2011. "Non-Suicidal Self-Injury in United States Adults: Prevalence, Sociodemographics, Topography, and Functions." *Psychological Medicine* 41, no. 9: 1981–1986.
- Lei, H., J. Xiong, Y. Rao, T. Zhu, and X. Zhang. 2024. "Relationships Among Self-Esteem, Depression, and Self-Injury in Adolescents: A Longitudinal Study." *Frontiers in Public Health* 12: 1406283.
- Liang, K., L. Zhao, Y. Lei, et al. 2022. "Nonsuicidal Self-Injury Behaviour in a City of China and Its Association With Family Environment, Media Use, and Psychopathology." *Comprehensive Psychiatry* 115: 152311.
- Liu, J., Y. Gao, C. Liang, and X. Liu. 2022. "The Potential Addictive Mechanism Involved in Repetitive Nonsuicidal Self-Injury: The Roles of Emotion Dysregulation and Impulsivity in Adolescents." *Journal of Behavioral Addictions* 11, no. 4: 953–962.
- Liu, W., L. Chen, and X. Tu. 2017. "Chinese Adaptation of Emotion Regulation Questionnaire for Children and Adolescents (ERQ-CCA): A Psychometric Evaluation in Chinese Children." *International Journal of Psychology* 52, no. 5: 398–405.
- Ma, S., Y. Wan, S. Zhang, et al. 2018. "Mediating Effect of Psychological Symptoms, Coping Styles, and Impulsiveness on the Relationship Between Childhood Abuses and Non-Suicidal Self-Injuries Among Middle School Students." *Wei Sheng Yan Jiu, Journal of Hygiene Research* 47, no. 4: 530–535.
- Martin, J., K. L. Raby, M. H. Labella, and G. I. Roisman. 2017. "Childhood Abuse and Neglect, Attachment States of Mind, and Non-Suicidal Self-Injury." *Attachment & Human Development* 19, no. 5: 425–446.
- Mehmood, A., T. Bu, E. Zhao, et al. 2021. "Exploration of the Psychological Mechanisms of Smartphone Addiction Among International Students of China by Selecting the Framework of the I-PACE Model." *Frontiers in Psychology* 12: 758610.
- Misiak, B., M. Szewczuk-Bogusławska, J. Samochowiec, A. A. Moustafa, and Ł. Gawęda. 2023. "Unraveling the Complexity of Associations Between a History of Childhood Trauma, Psychotic-Like Experiences, Depression, and Non-Suicidal Self-Injury: A Network Analysis." *Journal of Affective Disorders* 337: 11–17.
- Najmi, S., D. M. Wegner, and M. K. Nock. 2007. "Thought Suppression and Self-Injurious Thoughts and Behaviors." *Behaviour Research and Therapy* 45, no. 8: 1957–1965.
- Nock, M. K., and C. B. Cha. 2009. "Psychological Models of Nonsuicidal Self-Injury".
- Podsakoff, P. M., S. B. MacKenzie, J.-Y. Lee, and N. P. Podsakoff. 2003. "Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies." *Journal of Applied Psychology* 88, no. 5: 879–903.
- Qu, D., Y. Wang, Z. Zhang, et al. 2022. "Psychometric Properties of the Chinese Version of the Functional Assessment of Self-Mutilation (FASM) in Chinese Clinical Adolescents." *Frontiers in Psychiatry* 12: 755857.
- Robbins, T. W., and L. Clark. 2015. "Behavioral Addictions." *Current Opinion in Neurobiology* 30: 66–72.
- Sadeh, N., E. A. Londahl-Shaller, A. Piatigorsky, et al. 2014. "Functions of Non-Suicidal Self-Injury in Adolescents and Young Adults With Borderline Personality Disorder Symptoms." *Psychiatry Research* 216, no. 2: 217–222.
- Selby, E. A., J. Franklin, A. Carson-Wong, and S. L. Rizvi. 2013. "Emotional Cascades and Self-Injury: Investigating Instability of Rumination and Negative Emotion." *Journal of Clinical Psychology* 69, no. 12: 1213–1227.
- Southwick, S. M., and D. S. Charney. 2012. "The Science of Resilience: Implications for the Prevention and Treatment of Depression." *Science* 338, no. 6103: 79–82.
- Stanciu, D., and A. Calugar. 2022. "What Is Irrational in Fearing Missing Out on Being Online? An Application of the I-PACE Model Regarding the Role of Maladaptive Cognitions in Problematic Internet Use." *Computers in Human Behavior* 135: 107365.
- Victor, S. E., J. J. Muehlenkamp, N. A. Hayes, G. J. Lengel, D. M. Styer, and J. J. Washburn. 2018. "Characterizing Gender Differences in Nonsuicidal Self-Injury: Evidence From a Large Clinical Sample of Adolescents and Adults." *Comprehensive Psychiatry* 82: 53–60.
- Wang, Y.-P., and C. Gorenstein. 2013. "Psychometric Properties of the Beck Depression Inventory-II: A Comprehensive Review." *Brazilian Journal of Psychiatry* 35: 416–431.

- Weedage, D., N. Kool-Goudzwaard, D. Meijncens, R. R. Vermeiren, and C. Boonmann. 2025. "Resilience Revisited: A Systematic Review and Synthesis of Non-Suicidal Self-Injury (NSSI) and Its Relation With Resilience." *BMC Psychiatry* 25, no. 1: 1–19.
- Xavier, A., J. Pinto-Gouveia, M. Cunha, and A. Dinis. 2017. "Longitudinal Pathways for the Maintenance of Non-Suicidal Self-Injury in Adolescence: The Pernicious Blend of Depressive Symptoms and Self-Criticism." Paper presented at the Child & Youth Care Forum.
- Xiao, Q., X. Song, L. Huang, D. Hou, and X. Huang. 2022. "Global Prevalence and Characteristics of Non-Suicidal Self-Injury Between 2010 and 2021 Among a Non-Clinical Sample of Adolescents: A Meta-Analysis." *Frontiers in Psychiatry* 13: 912441.
- Yang, H., Z. Wang, J. Song, et al. 2020. "The Positive and Negative Rumination Scale: Development and Preliminary Validation." *Current Psychology* 39: 483–499.
- Yu, X., and J. Zhang. 2007. "Factor Analysis and Psychometric Evaluation of the Connor-Davidson Resilience Scale (CD-RISC) With Chinese People." *Social Behavior and Personality: An International Journal* 35, no. 1: 19–30.
- Zashchirinskaia, O., and E. Isagulova. 2023. "Childhood Trauma as a Risk Factor for High-Risk Behaviors in Adolescents With Borderline Personality Disorder." *Iranian Journal of Psychiatry* 18, no. 1: 65–71.
- Zhang, B., W. Zhang, L. Sun, C. Jiang, Y. Zhou, and K. He. 2023. "Relationship Between Alexithymia, Loneliness, Resilience, and Non-Suicidal Self-Injury in Adolescents With Depression: A Multi-Center Study." *BMC Psychiatry* 23, no. 1: 445.
- Zhao, X., Y. Zhang, L. Li, and Y. Zhou. 2005. "Evaluation of the Reliability and Validity of the Chinese Version of the Childhood Trauma Questionnaire." *Chinese Journal of Tissue Engineering Research*: 209–211.
- Zhou, Y., N. X. Yu, Z. Liang, S. Feng, and Q. Zhang. 2024. "Preadolescents' Executive Functions and Resilience Development: A Cascade Model of Resilience Resources and Resilient Functioning." *Journal of Adolescence* 96, no. 8: 1929–1941.
- Zhu, J., R. Qian, H. Zhong, Y. Li, X. Liu, and J. Ma. 2022. "Factors Influencing the Addiction Characteristics of Non-Suicidal Self-Injurious Behaviors in Adolescents: A Case-Control Study." *Frontiers in Psychiatry* 13: 1033242.
- Zhuo, C., G. Chen, C. Lin, et al. 2022. "A Borderline Personality Assessment for Adolescents: Validity and Reliability of the Chinese Language Borderline Personality Features Scale (Short Form Version) for Adolescents/Children." *Frontiers in Psychiatry* 13: 1050559.